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	Unit	Reference	Clause	Symbol	Formula	TREFOIL GND	TREFOIL DUCT	TREFOIL AIR
DESIGN DATA								
Bonding Type		Design data				SOLID	SOLID	SOLID
Cond Mat		Design data				Al	Al	Al
Cond size	mm2	Design data				185	185	185
Cond design		Design data			cc	stranded	stranded	stranded
Cond Dn	mm	Design data		Dc		15.9	15.9	15.9
Cond Sc Dn	mm	Design data		Ds		17.6	17.6	17.6
Ins Mat		Design data				XLPE	XLPE	XLPE
Ins Dn	mm	Design data		Di		33.9	33.9	33.9
Ins Sc Dn	mm	Design data		dx		35.6	35.6	35.6
Ins Tn	mm	Design data		Ti		8.15	8.15	8.15
Cable Dn	mm	Design data		De		43.4	43.4	43.4
Max cond temp	C	Design Data		Tmax		90	90	90
Rated voltage	kV	IEC60502-2	Table 1	U		33	33	33
Earth voltage	kV	IEC60502-2	Table 1	Uo		19.05	19.05	19.05
PARAMETER SETTING								
Soil TR	K.m/W	Input		p		1.20	1.20	--
Frequency	Hz	Input		f		50	50	50
Ambient temp.	C	Input		Ta		30	30	40
Depth	mm	Input		L		1000	1000	--
Duct Type		Input				--	HDPE	--
External diam of duct	mm	Input		dc		--	110.0	--
Thickness of duct	mm	Input		Td		--	5.0	--
Temp rise, amb to max.	C	IEC60287-1-1	1.4.1.1	dT	Tmax - Ta	60	60	50
Core temperature, Tc	C	Iteration		Tc		90.00	90.00	90.00
Ins Relative permittivity		IEC60287-1-1	Table 3	er		2.500	2.500	2.500
Loss factor		IEC60287-1-1	Table 3	tand		0.0010	0.0010	0.0010
Dielectric loss	W/m/ph	EC60287-1-1	2.2	Wd	wCUo2tand	0.0242	0.0242	0.0242
DC RESISTANCE								
DC resistant at 20C	O/m	IEC60228	Class 2	Ro		1.640E-04	1.640E-04	1.640E-04
DC resistant at ambient	O/m	IEC60287-1-1	2.1.1	Ro amb	Ro(1+a20(Ta-20))	1.706E-04	1.706E-04	1.706E-04
DC resistant at 90C	O/m	IEC60287-1-1	2.1.1	Ro max	Ro(1+a20(Tmax-20))	2.103E-04	2.103E-04	2.103E-04
AC RESISTANCE								
Strand constant		IEC60287-1-1	Table 2	ks		1.000	1.000	1.000

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Strand constant		IEC60287-1-1	Table 2	kp		0.800	0.800	0.800
Bessel function		IEC60287-1-1	2.1.2	Xs4	[ks(8pif/Ro)10-7]2	0.357	0.357	0.357
Skin effect		IEC60287-1-1	2.1.2	Ys	Xs4/(192+0.8 Xs4)	0.002	0.002	0.002

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Proximity function		IEC60287-1-1	2.1.4	Xp4	[kp(8pif/Ro)10-7]2	0.229	0.229	0.229
Proximity function		IEC60287-1-1	2.1.4	Yp"	Xp4/(192+0.8 Xp4)	0.001	0.001	0.001
Proximity effect		IEC60287-1-1	2.1.4	Yp	Yp"(Dc/s)2[0.312(Dc/s)2+1.18/(Yp"+0.27)]	0.001	0.000	0.001
AC resistant at 90C	O/m	IEC60287-1-1	2.1	R90	Ro max(1+Ys+Yp)	2.108E-04	2.107E-04	2.108E-04
SHEATH LOSS FACTOR								
Equivalent mean diameter	mm	Design data		ds		36.0	36.0	36.0
Equivalent resistant at 20C	O/m	IEC60287-1-1		R20	p/A	3.448E-04	3.448E-04	3.448E-04
Equivalent resistant at Tc	O/m	IEC60287-1-1		Rs	Rs0(1+a20(Tc-20))	4.273E-04	4.285E-04	4.209E-04
Sheath Reactance	O/m	IEC60287-1-1	2.3.3	X	2wln(2s/d)x10-7	5.472E-05	1.137E-04	5.472E-05
Circulating current loss factor		IEC60287-1-1	2.3.1	l1'	Rs/R/(1+(Rs/X)2)	0.0327	0.1338	0.0332
Eddy current loss factor		IEC60287-1-1	2.3.6.1	l1"	(Rs/R)gslo(1+D1+D2)+(b1ts)4/(12x1010)	0.0000	0.0000	0.0000
Sheath loss factor		IEC60287-1-1	2.3	l1	l1'+l1"	0.0327	0.1338	0.0332
THERMAL RESISTANCE								
T1 factor		IEC60287-2-1				1.000	1.000	1.000
Thermal Resistant Insulation	K.m/W	IEC60287-2-1		T1	(p/2p)LN[1+(2t/dc)]	0.451	0.451	0.451
T3 factor		IEC60287-2-1				1.600	1.000	1.000
Thermal Resistant Outercovering	K.m/W	IEC60287-2-1		T3	(p/2p)LN[1+(2t/D)]	0.208	0.130	0.130
T4' air gap	K.m/W	IEC60287-2-1	2.2.7	T4'	U/(1+0.1(V+YTm)De)	--	0.5288	--
T4" duct wall	K.m/W	IEC60287-2-1	2.2.7	T4"	(pT/2p)LN(Do/Dd)	--	0.0531	--
T4"' soil	K.m/W	IEC60287-2-1	2.2.7	T4"'	Tself+2xTmutual	--	1.7946	--
Thermal Resistant External	K.m/W	IEC60287-2-1	2.2	T4	(pT/2p)LN(u+SQRT(u2-1))	2.231	2.377	1.011
CURRENT RATING								
Critical Temp Rise of Soil	C	IEC60287-1-1		dT	Wk(Pc/2p)LN(Gf)	0.0	0.0	0.0
Top Equation		IEC60287-1-1	1.4.1	Etop	DT-Wd(0.5T1+n(T2+T3+T4))-dT	5.994E+01	5.993E+01	4.997E+01
Bottom Equation		IEC60287-1-1	1.4.1	Ebot	RT1+nR(1+l1+l2)(T3+T4)	6.262E-04	6.939E-04	3.438E-04
Calculated Amps	A	IEC60287-1-1	1.4.1	I	v(Etop/Ebot)	309	294	381

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Calculated MVA	MVA	IEC60287-1-1	1.4.1	MVA	v3UI	17.7	16.8	21.8
ELECTRICAL DATA								
Inductance	uH/m	IEC60287-1-1	2.3	L	0.2LN(2s/Dm)	0.174	0.174	0.174
Reactance	uO/m	IEC60287-1-1	2.3	X	w Lm x10-3	54.7	54.7	54.7
Pos Seq Resistance	uO/m			R1	R'1(1+l1+l2)	217.7	217.7	217.7
Pos Seq Impedance	uO/m			Z1	v(R12+X12)	224.5	224.5	224.5
Capacitance	uF/km	IEC60287-1-1	2.2	C	e/(18LN(Di/ds))	0.212	0.212	0.212
THERMAL LOSS								

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Thermal Resistant Ins	K.m/W	IEC60287-2-1		T1	(p/2p)LN[1+(2t/dc)]	0.451	0.451	0.451
Thermal Resistant Osh	K.m/W	IEC60287-2-1		T3	(p/2p)LN[1+(2t/D)]	0.208	0.130	0.130
Thermal Resistant Ext	K.m/W	IEC60287-2-1	2.2	T4	(pT/2p)LN(u+SQRT(u2-1))	2.231	2.377	1.011
Thermal Resistant Total	K.m/W	Mc Graw hill	Page 34	ST	T1+T2+T3+T4	2.891	2.958	1.593
Temp Over Ins	C	Mc Graw hill	Page 34	T1	T1=W*T1	9.42	9.33	14.30
Temp Over Osh	C	Mc Graw hill	Page 34	T3	T3=W*T3	46.54	49.09	32.04
Temp Ambient	C	Mc Graw hill	Page 34	Ti	Ta=W*T4	30	30	40
Losses - Conductor	W/m	IEC60287-1-1	2.4.2	Wc	3I2R	20.18	18.20	30.64
Losses - Dielectric	W/m	IEC60287-1-1	2.2	Wd	3wCUo2tand	0.024	0.024	0.024
Losses - Metal sheath	W/m	IEC60287-1-1		Wm	Wcl1	0.66	2.44	1.02